

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("C:/Users/admin/Desktop/C03099/Iris (1).csv")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Id                    150 non-null   int64
1   SepalLengthCm         150 non-null   float64
2   SepalWidthCm           150 non-null   float64
3   PetalLengthCm          150 non-null   float64
4   PetalWidthCm           150 non-null   float64
5   Species                150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	
Species						
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
stats = df.describe()
```

```
stats.loc['variance'] = df.var(numeric_only=True)
```

```
quantiles = df.quantile([0.1, 0.9], numeric_only=True)
```

```
stats
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	\
count	150.000000	150.000000	150.000000	150.000000	
mean	75.500000	5.843333	3.054000	3.758667	
std	43.445368	0.828066	0.433594	1.764420	
min	1.000000	4.300000	2.000000	1.000000	
25%	38.250000	5.100000	2.800000	1.600000	

50%	75.500000	5.800000	3.000000	4.350000
75%	112.750000	6.400000	3.300000	5.100000
max	150.000000	7.900000	4.400000	6.900000
variance	1887.500000	0.685694	0.188004	3.113179

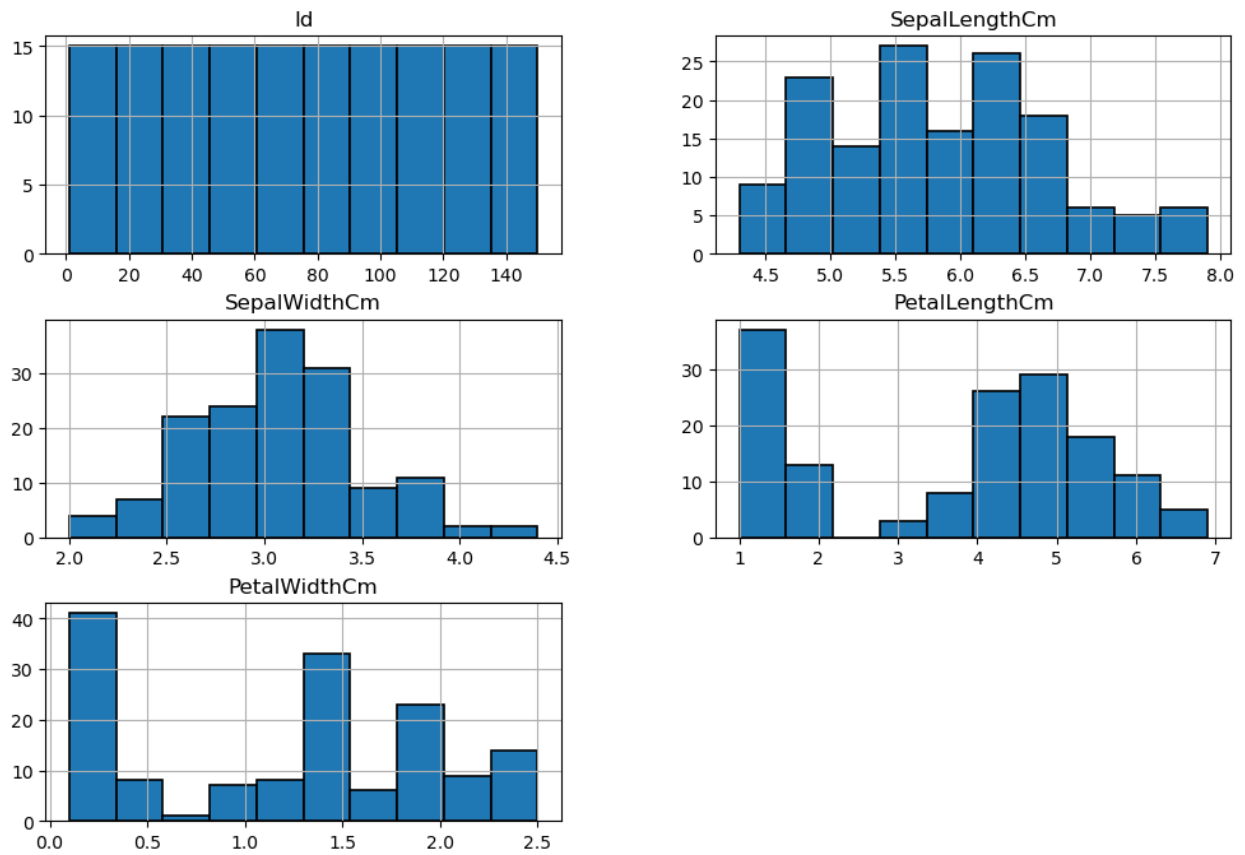
	PetalWidthCm
count	150.000000
mean	1.198667
std	0.763161
min	0.100000
25%	0.300000
50%	1.300000
75%	1.800000
max	2.500000
variance	0.582414

quantiles

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0.1	15.9	4.8	2.50	1.4	0.2
0.9	135.1	6.9	3.61	5.8	2.2

```
df.hist(edgecolor='black', linewidth=1.2, figsize=(12, 8))
plt.suptitle("Histograms of Iris Features")
plt.show()
```

Histograms of Iris Features



```
plt.figure(figsize=(10, 6))
df.boxplot()
plt.title("Boxplot of Iris Features")
plt.ylabel("Value in cm")
plt.show()
```

